



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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29 May, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
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Executive Summary

Executive Summary:

This project aims to predict whether SpaceX's Falcon 9 first stage will land successfully after launch. Using publicly available SpaceX data, we:

- Collected data via SpaceX REST API and web scraping
- Performed data wrangling and exploratory data analysis
- Built interactive maps with Folium and dashboards with Plotly Dash
- Developed classification models to predict landing success
- Achieved [X]% accuracy using [best model name]

Key Findings:

- Launch success rate has improved significantly since 2013
- KSC LC-39A has the highest success rate among launch sites
- Payload mass between 4000-6000 kg shows higher success rates
- Orbit type ES-L1 and SSO have lower success rates

Introduction

Project Background and Context:

- SpaceX advertises Falcon 9 rocket launches at \$62 million
- Competitors charge around \$165 million per launch
- The lower cost comes from reusing the first stage
- Predicting landing success helps estimate launch costs

Problems to Answer:

1. What factors affect Falcon 9 first stage landing success?
2. Which launch sites have the highest success rates?
3. How does payload mass affect landing outcome?
4. Which orbit types have better success rates?
5. Can we predict landing success using classification models?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

- Data Collection Methodology
- The project follows a complete data science pipeline from data collection to predictive modeling.

Source 1: SpaceX REST API

The SpaceX API provides structured data about all past launches.

Data Retrieved:

- Launch date and flight number
- Rocket name and booster version
- Launch site coordinates and name
- Payload mass, customer, and orbit
- Landing outcome and landing type

- **Source 2: Web Scraping (Wikipedia)**
- Additional data scraped from Wikipedia for verification and completeness.

Aspect	Details
URL	Wikipedia SpaceX launch page
Method	BeautifulSoup HTML parsing
Data	Launch tables with outcomes

Data Collection

- **Data Collection Process**
- **1. API Data Collection**
 - Sent HTTP requests using Python requests
 - Retrieved launch records from the SpaceX API
 - Converted JSON responses into Pandas DataFrames
- **Key Tools**
 - Python
 - Requests
 - Pandas
- **Workflow**
 - SpaceX API → HTTP Request → JSON Response → Pandas DataFrame
 -

Web Scraping from Wikipedia

- Requested Falcon 9 launch web page
- Parsed HTML using BeautifulSoup
- Extracted launch tables and column values
- Cleaned and stored data in structured format

• Key Tools

- BeautifulSoup
- Requests
- Pandas

• Workflow

- Wikipedia Page → HTML Parsing → Extract Tables → Clean Data → DataFrame

- **3. SQL Data Extraction**

- Queried launch datasets using SQL
- Retrieved filtered and aggregated launch information
- Used SQL for exploratory data analysis

- **Key Tools**

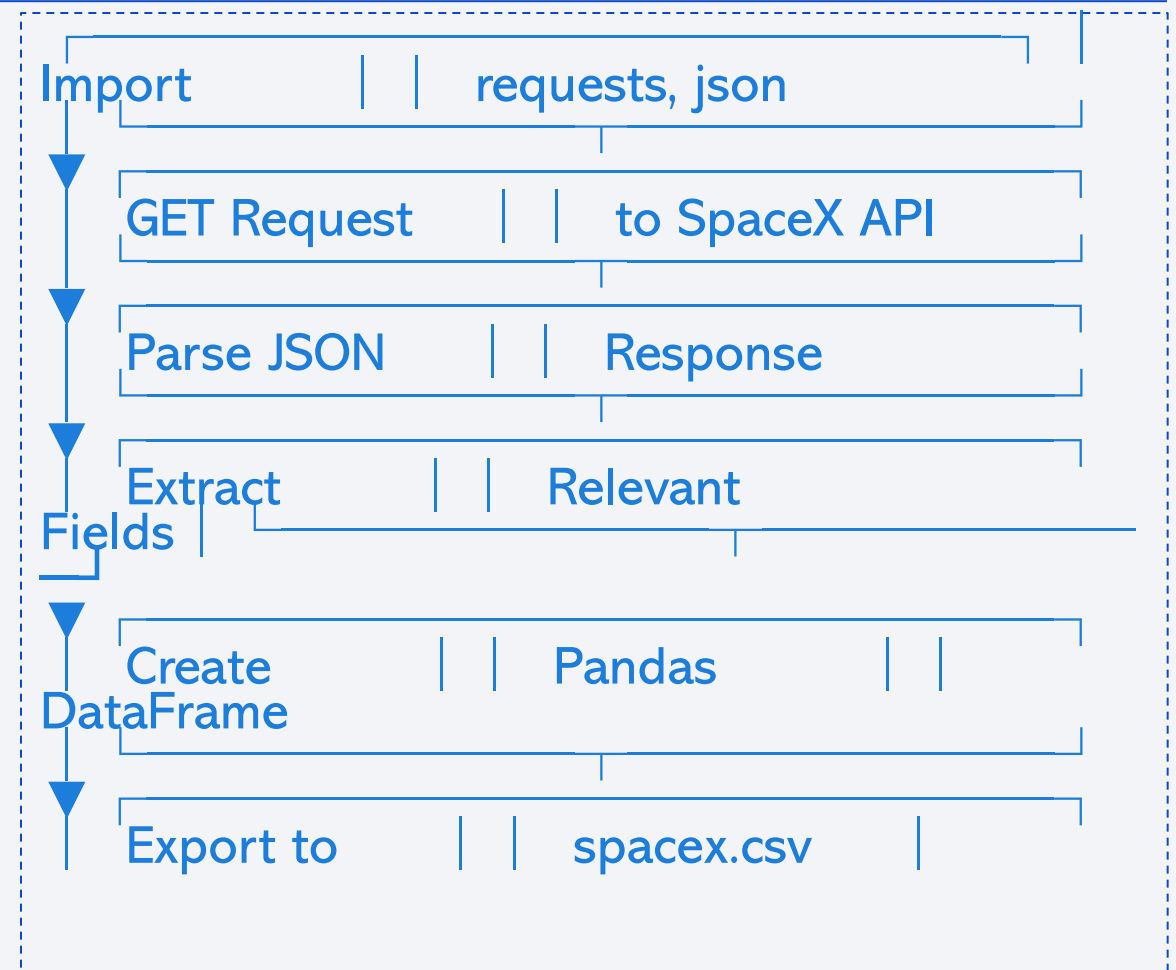
- SQL
- SQLite / IBM Db2

- **Workflow**

- Database → SQL Queries → Filtered Results → Analysis

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook (must include completed code cell and outcome cell), as an external reference and peer-review purpose



Data Collection - Scraping

- **Web Scraping Process**

- Requested Falcon 9 launch webpage using HTTP requests
- Parsed HTML content using BeautifulSoup
- Extracted launch tables and records
- Cleaned and transformed data into Pandas DataFrames

- **Key Tools**

- Python
- Requests
- BeautifulSoup
- Pandas

- **GitHub Repository:**

<https://github.com/fariha-codes/TestRepo/blob/main/jupyter-labs-webscraping.ipynb>

Wikipedia Falcon 9 Page



HTTP Request



HTML Response



BeautifulSoup Parsing



Extract HTML Tables



Clean & Process Data



Pandas DataFrame

Data Wrangling

- **Data Wrangling Process**

- Removed unnecessary columns
- Handled missing and null values
- Converted data types into proper formats
- Extracted useful variables from raw data
- Encoded categorical variables
- Standardized and cleaned dataset for analysis

- **Key Tools**

- Pandas
- NumPy
- Python
- [https://github.com/fariha-codes/TestRepo/blob/main/labs-jupyter-spacex-Data%20wrangling-v2%20\(1\).ipynb](https://github.com/fariha-codes/TestRepo/blob/main/labs-jupyter-spacex-Data%20wrangling-v2%20(1).ipynb)

EDA with Data Visualization

- Visualization techniques used:
 - Scatter plots
 - Bar charts
 - Line charts
 - Correlation analysis
- EDA helped identify relationships between payload, orbit type, launch site, and landing success.

EDA with SQL

- SQL analysis performed:
- Launch site analysis
- Payload calculations
- Booster performance evaluation
- Landing outcome ranking
- Success and failure statistics

Build an Interactive Map with Folium

- Interactive maps were created to:
- Visualize launch sites
- Analyze landing outcomes
- Calculate proximity to coastlines, highways, and railways
- Display launch site markers and success indicators

Build a Dashboard with Plotly Dash

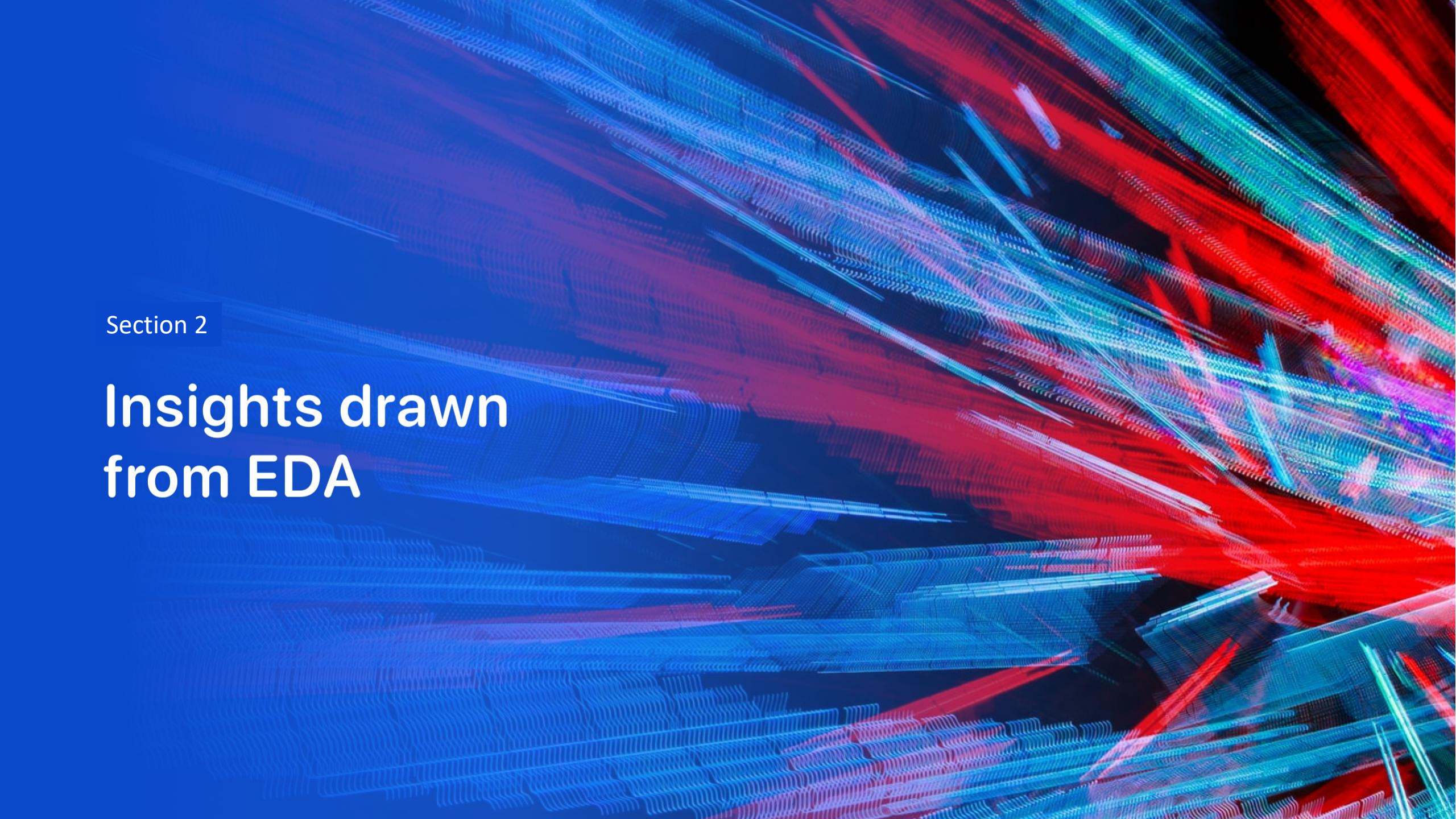
- An interactive dashboard was developed with:
 - Pie charts for launch success counts
 - Scatter plots for payload analysis
 - Interactive filters and sliders
 - Launch site comparison tools

Predictive Analysis (Classification)

- Classification models used:
 - Logistic Regression
 - Support Vector Machine (SVM)
 - Decision Tree
 - K-Nearest Neighbors (KNN)
- Model evaluation metrics:
 - Accuracy
 - Confusion Matrix

Results

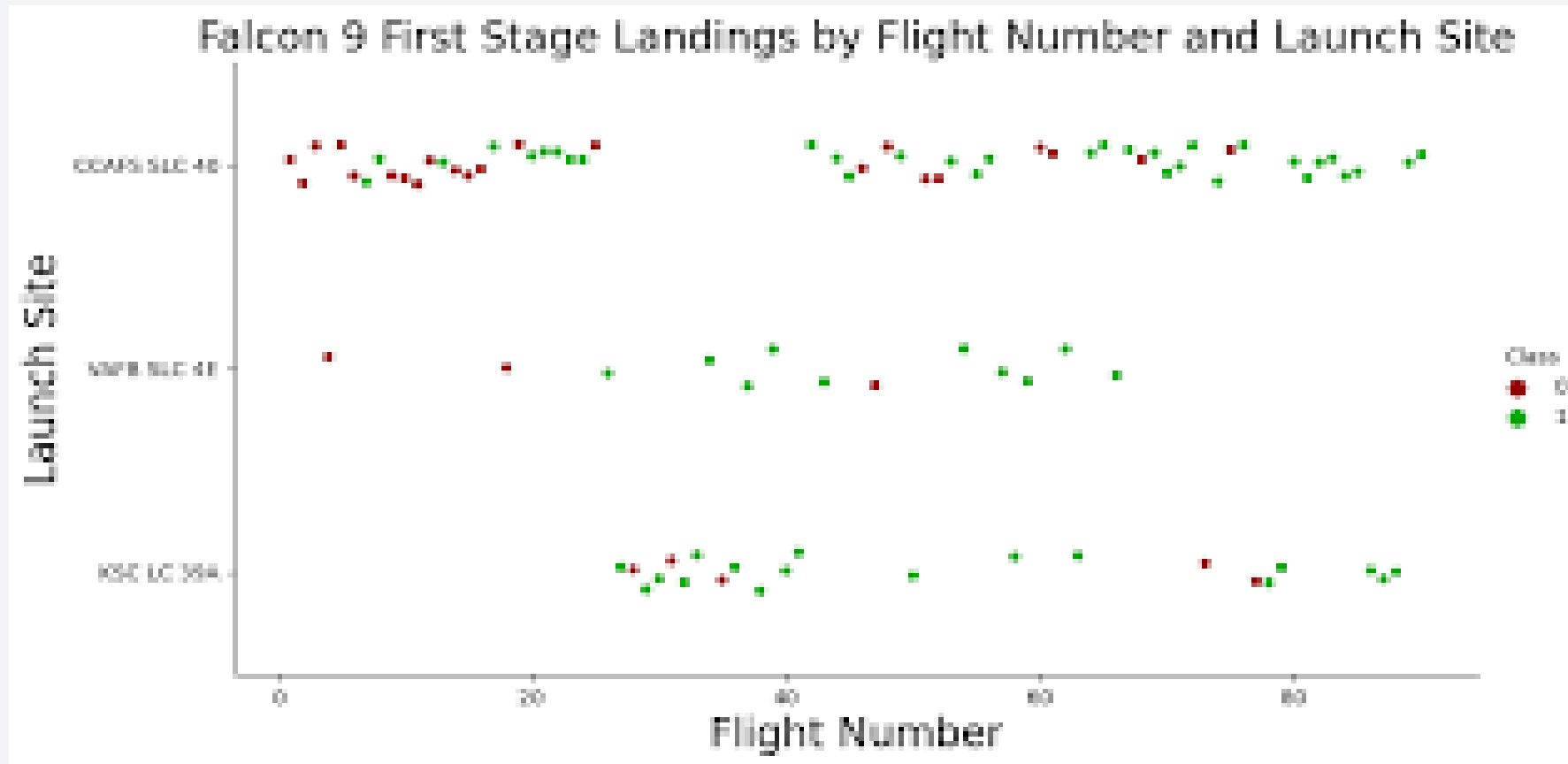
- Key findings:
 - Launch success increased over time
 - Certain orbit types had higher success rates
 - Payload mass influenced landing outcomes
 - Machine learning models successfully predicted landing success

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site



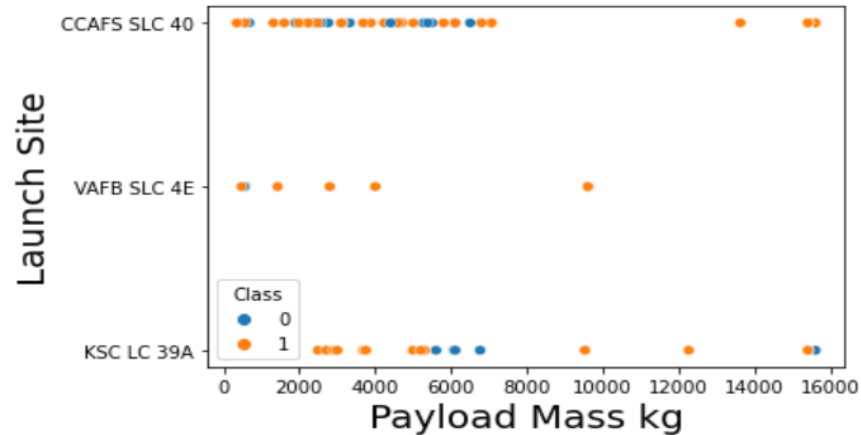
Payload vs. Launch Site

TASK 2: Visualize the relationship between Payload and Launch Site

We also want to observe if there is any relationship between launch sites and their payload mass.

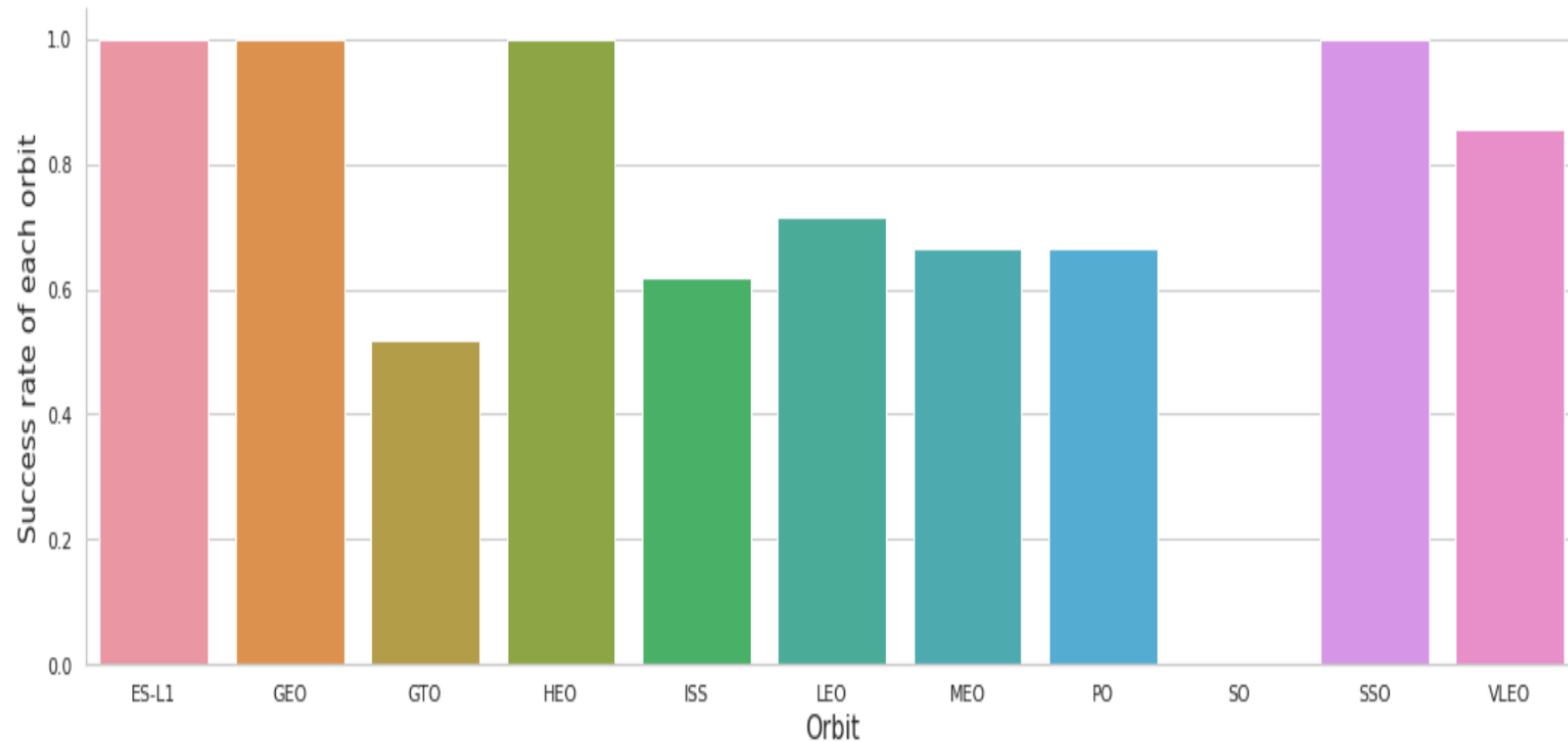
In [8]:

```
# Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the Launch site, and hue to be the
sns.scatterplot(x='PayloadMass',y='LaunchSite', hue='Class', data=df)
plt.xlabel("Payload Mass kg", fontsize=20)
plt.ylabel("Launch Site", fontsize=20)
plt.show()
```



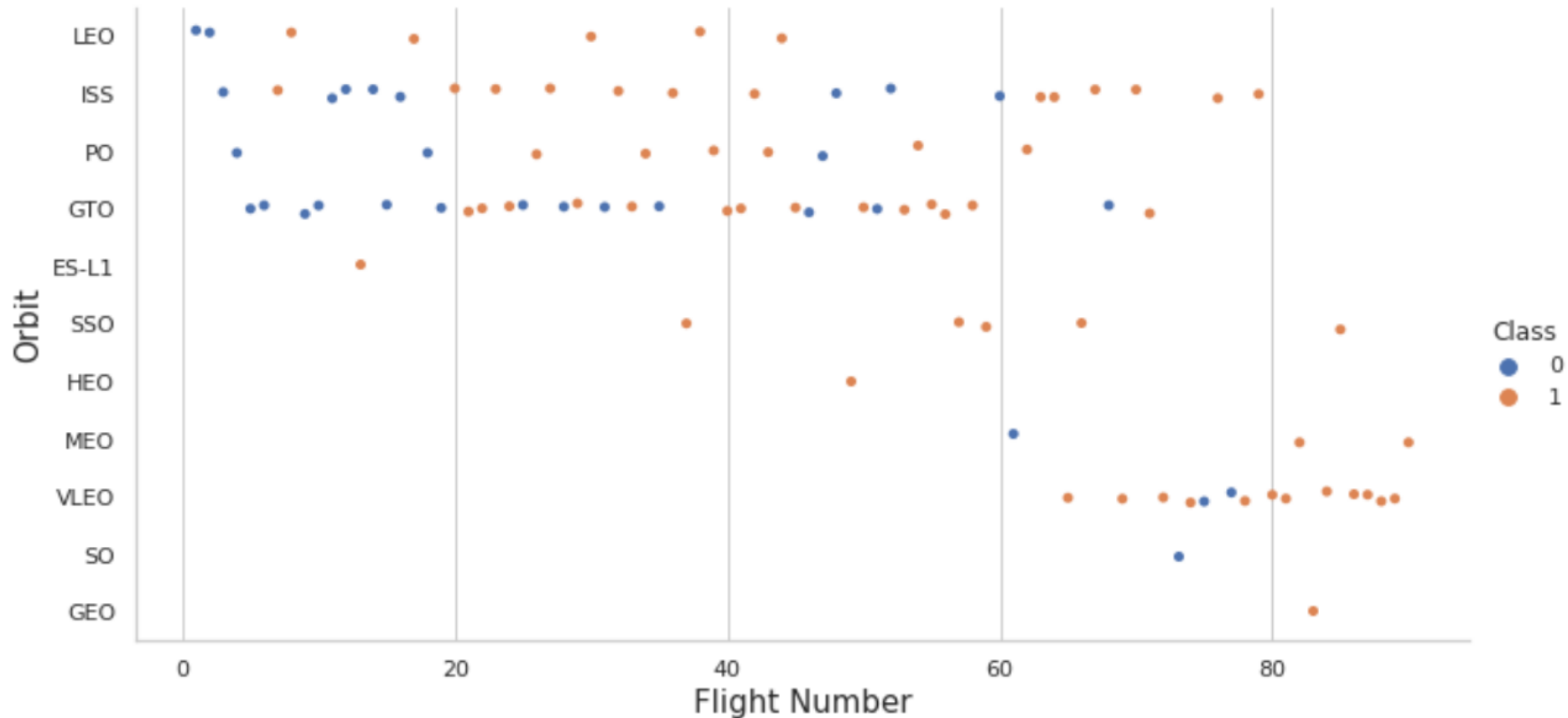
Now if you observe Payload Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Success Rate vs. Orbit Type



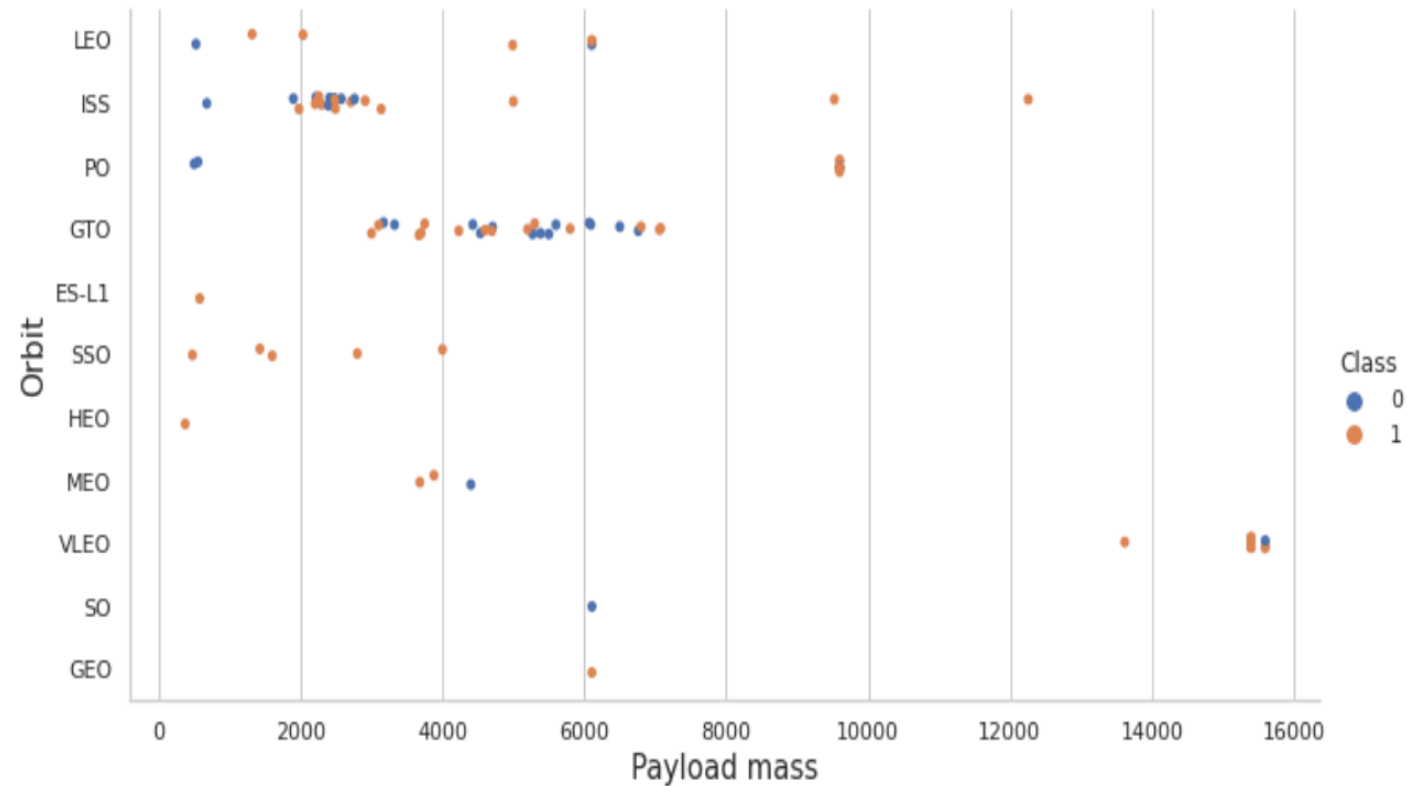
Analyze the plotted bar chart try to find which orbits have high sucess rate.

Flight Number vs. Orbit Type



You should see that in the LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

Payload vs. Orbit Type

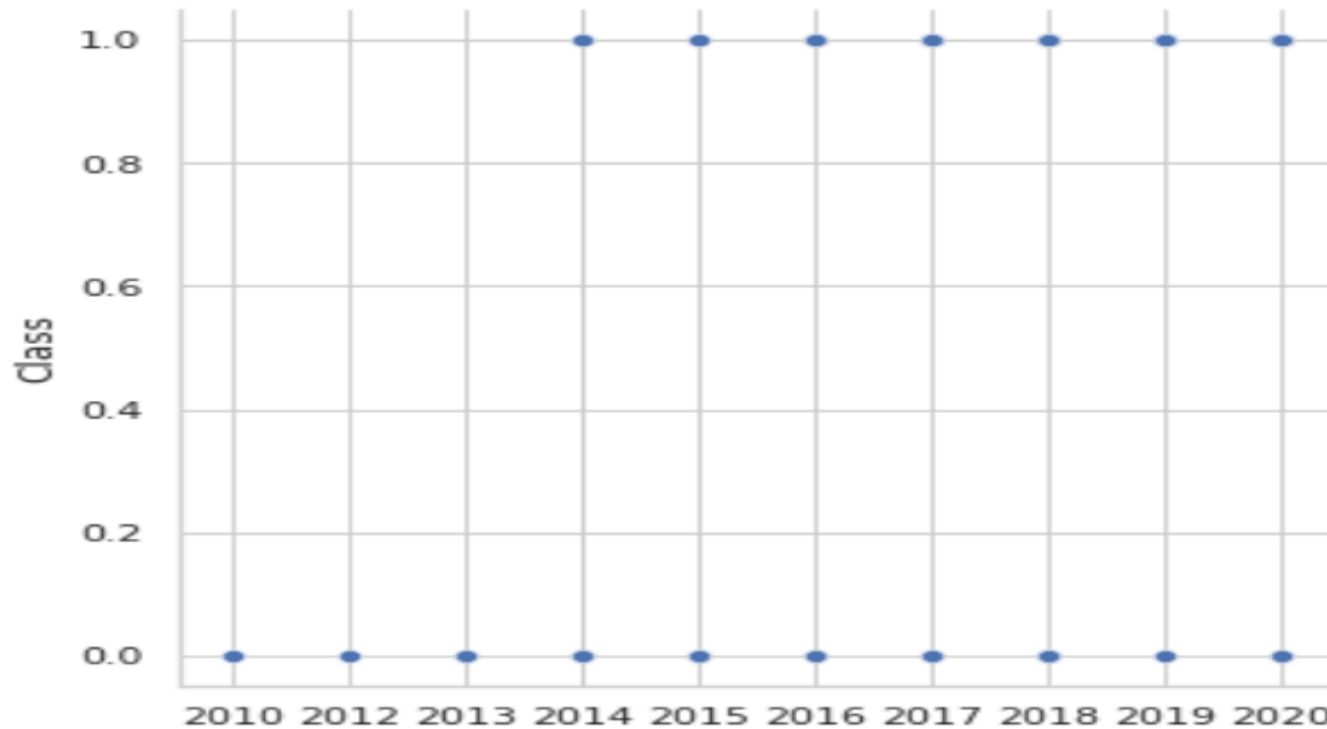


With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.

However for GTO we cannot distinguish this well as both positive landing rate and negative landing (unsuccessful mission) are both there here.

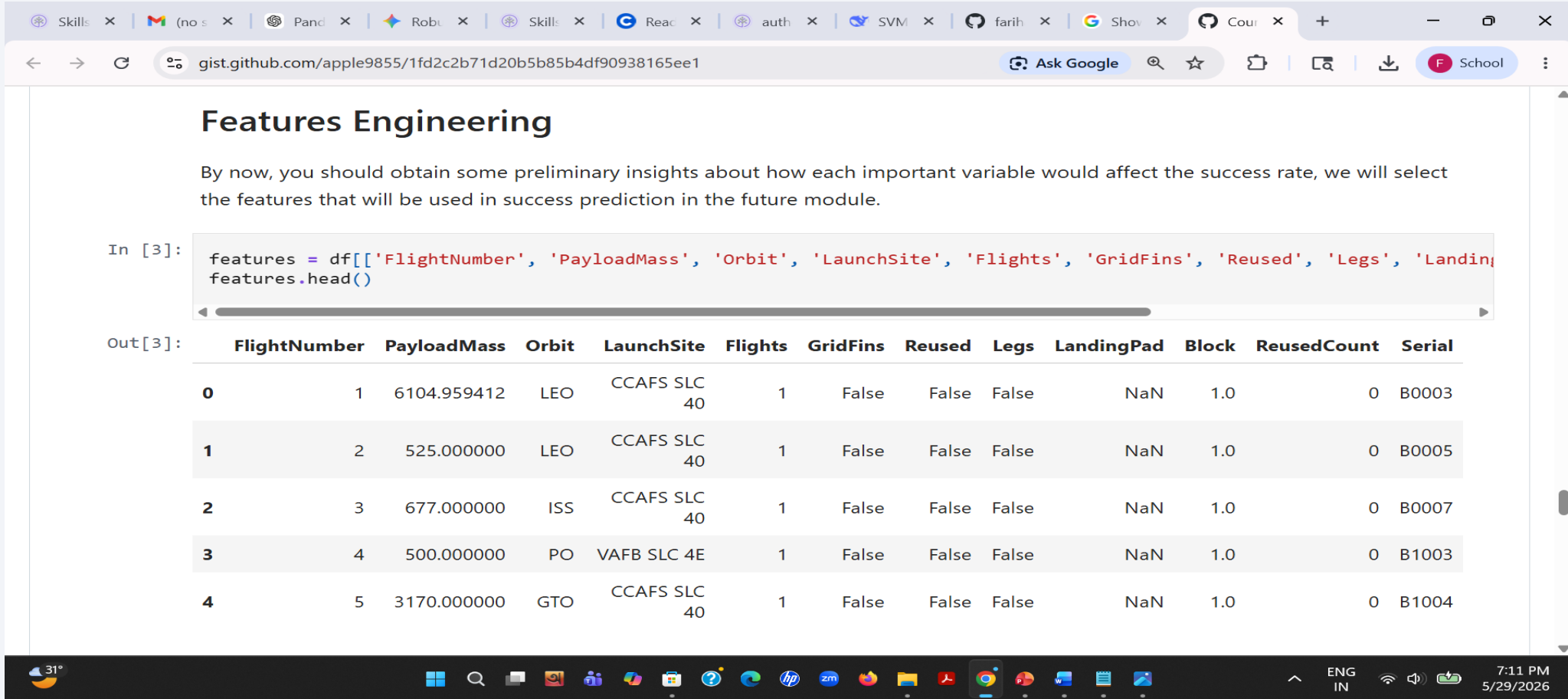
Launch Success Yearly Trend

Out[54]: <seaborn.axisgrid.FacetGrid at 0x7f57b888a280>



you can observe that the success rate since 2013 kept increasing till 2020

All Launch Site Names



The screenshot shows a web browser window displaying a Jupyter Notebook. The browser's address bar shows a GitHub gist URL. The notebook has a title "Features Engineering" and a paragraph explaining the selection of features for success prediction. Below this, a code cell shows the selection of specific features from a DataFrame. The output cell displays the first five rows of the selected features as a table.

Features Engineering

By now, you should obtain some preliminary insights about how each important variable would affect the success rate, we will select the features that will be used in success prediction in the future module.

In [3]:

```
features = df[['FlightNumber', 'PayloadMass', 'Orbit', 'LaunchSite', 'Flights', 'GridFins', 'Reused', 'Legs', 'LandingPad', 'Block', 'ReusedCount', 'Serial']]
features.head()
```

Out[3]:

	FlightNumber	PayloadMass	Orbit	LaunchSite	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial
0	1	6104.959412	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0003
1	2	525.000000	LEO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0005
2	3	677.000000	ISS	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B0007
3	4	500.000000	PO	VAFB SLC 4E	1	False	False	False	NaN	1.0	0	B1003
4	5	3170.000000	GTO	CCAFS SLC 40	1	False	False	False	NaN	1.0	0	B1004

Launch Site Names Begin with 'CCA'

```
In [11]: task_2 = '''
          SELECT *
          FROM SpaceX
          WHERE LaunchSite LIKE 'CCA%'
          LIMIT 5
          '''
          create_pandas_df(task_2, database=conn)
```

```
Out[11]:
```

	date	time	boosterversion	launchsite	payload	payloadmasskg	orbit	customer	missionoutcome	landingoutcome
0	2010-04-06	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	2010-08-12	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	2012-08-10	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
4	2013-01-03	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- We calculated the total payload carried by boosters from NASA as 45596 using the query below

Display the total payload mass carried by boosters launched by NASA (CRS)

```
In [12]: task_3 = '''
          SELECT SUM(PayloadMassKG) AS Total_PayloadMass
          FROM SpaceX
          WHERE Customer LIKE 'NASA (CRS)'
          '''
          create_pandas_df(task_3, database=conn)
```

```
Out[12]:
```

	total_payloadmass
0	45596

Average Payload Mass by F9 v1.1

Average Payload Mass by F9 v1.1

- We calculated the average payload mass carried by booster version F9 v1.1 as 2928.4

Display average payload mass carried by booster version F9 v1.1

```
In [13]: task_4 = '''
          SELECT AVG(PayloadMassKG) AS Avg_PayloadMass
          FROM SpaceX
          WHERE BoosterVersion = 'F9 v1.1'
          ...
          create_pandas_df(task_4, database=conn)
```

```
Out[13]:
```

	avg_payloadmass
0	2928.4

First Successful Ground Landing Date

First Successful Ground Landing Date

- We observed that the dates of the first successful landing outcome on ground pad was 22nd December 2015

```
In [14]: task_5 = '''
          SELECT MIN(Date) AS FirstSuccessfull_landing_date
          FROM SpaceX
          WHERE LandingOutcome LIKE 'Success (ground pad)'
          '''
          create_pandas_df(task_5, database=conn)
```

```
Out[14]:
```

	firstsuccessfull_landing_date
0	2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

```
In [15]: task_6 = '''
          SELECT BoosterVersion
          FROM SpaceX
          WHERE LandingOutcome = 'Success (drone ship)'
             AND PayloadMassKG > 4000
             AND PayloadMassKG < 6000
          ...
          create_pandas_df(task_6, database=conn)
```

```
Out[15]:
```

	boosterversion
0	F9 FT B1022
1	F9 FT B1026
2	F9 FT B1021.2
3	F9 FT B1031.2

- We used the **WHERE** clause to filter for boosters which have successfully landed on drone ship and applied the **AND** condition to determine successful landing with payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

```
List the total number of successful and failure mission outcomes

In [16]: task_7a = '''
          SELECT COUNT(MissionOutcome) AS SuccessOutcome
          FROM SpaceX
          WHERE MissionOutcome LIKE 'Success%'
          '''

          task_7b = '''
          SELECT COUNT(MissionOutcome) AS FailureOutcome
          FROM SpaceX
          WHERE MissionOutcome LIKE 'Failure%'
          '''

          print('The total number of successful mission outcome is:')
          display(create_pandas_df(task_7a, database=conn))
          print()
          print('The total number of failed mission outcome is:')
          create_pandas_df(task_7b, database=conn)

The total number of successful mission outcome is:
  successoutcome
0              100

The total number of failed mission outcome is:
Out[16]:  failureoutcome
0              1
```

- We used wildcard like '%' to filter for **WHERE** MissionOutcome was a success or a failure.

Boosters Carried Maximum Payload

- We determined the booster that have carried the maximum payload using a subquery in the **WHERE** clause and the **MAX()** function.

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
In [17]: task_8 = '''
          SELECT BoosterVersion, PayloadMassKG
          FROM SpaceX
          WHERE PayloadMassKG = (
                                SELECT MAX(PayloadMassKG)
                                FROM SpaceX
                                )
          ORDER BY BoosterVersion
          ...
          create_pandas_df(task_8, database=conn)
```

```
Out[17]:
```

	boosterversion	payloadmasskg
0	F9 B5 B1048.4	15600
1	F9 B5 B1048.5	15600
2	F9 B5 B1049.4	15600
3	F9 B5 B1049.5	15600
4	F9 B5 B1049.7	15600
5	F9 B5 B1051.3	15600
6	F9 B5 B1051.4	15600
7	F9 B5 B1051.6	15600
8	F9 B5 B1056.4	15600
9	F9 B5 B1058.3	15600
10	F9 B5 B1060.2	15600
11	F9 B5 B1060.3	15600

2015 Launch Records

- We used a combinations of the **WHERE** clause, **LIKE**, **AND**, and **BETWEEN** conditions to filter for failed landing outcomes in drone ship, their booster versions, and launch site names for year 2015

List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
In [18]: task_9 = '''
          SELECT BoosterVersion, LaunchSite, LandingOutcome
          FROM SpaceX
          WHERE LandingOutcome LIKE 'Failure (drone ship)'
          AND Date BETWEEN '2015-01-01' AND '2015-12-31'
          '''
          create_pandas_df(task_9, database=conn)
```

Out[18]:

	boosterversion	launchsite	landingoutcome
0	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
1	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Present your query result with a short explanation here

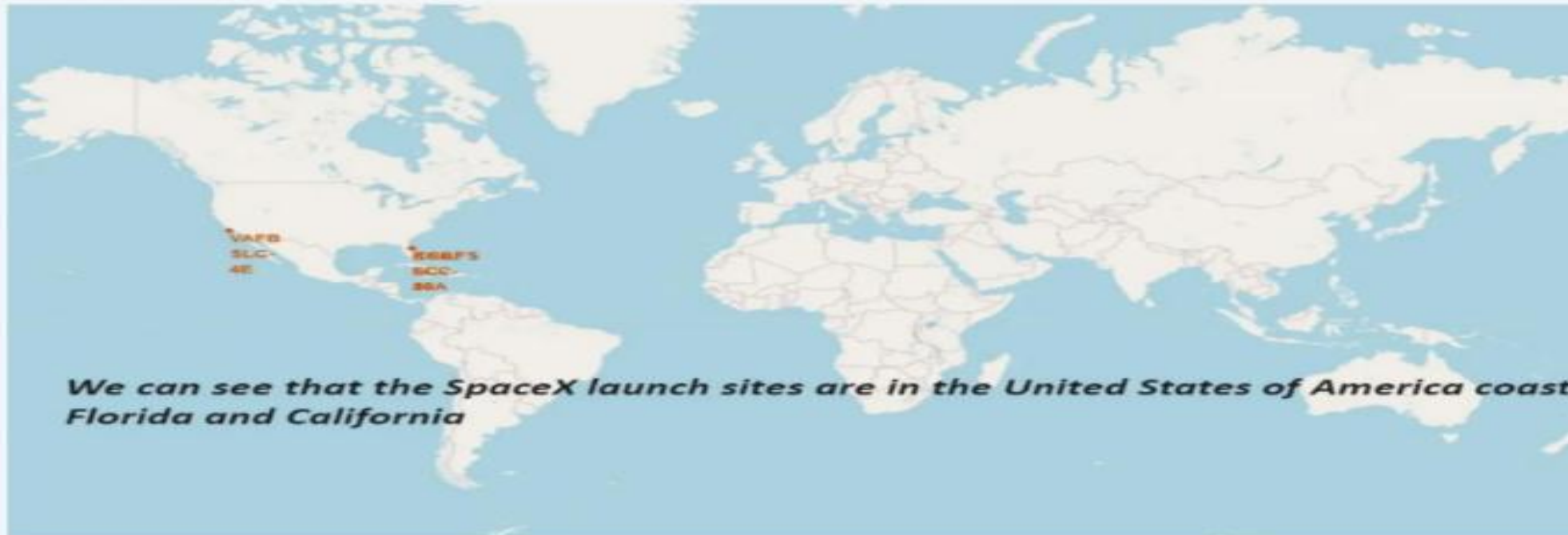
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

Section 3

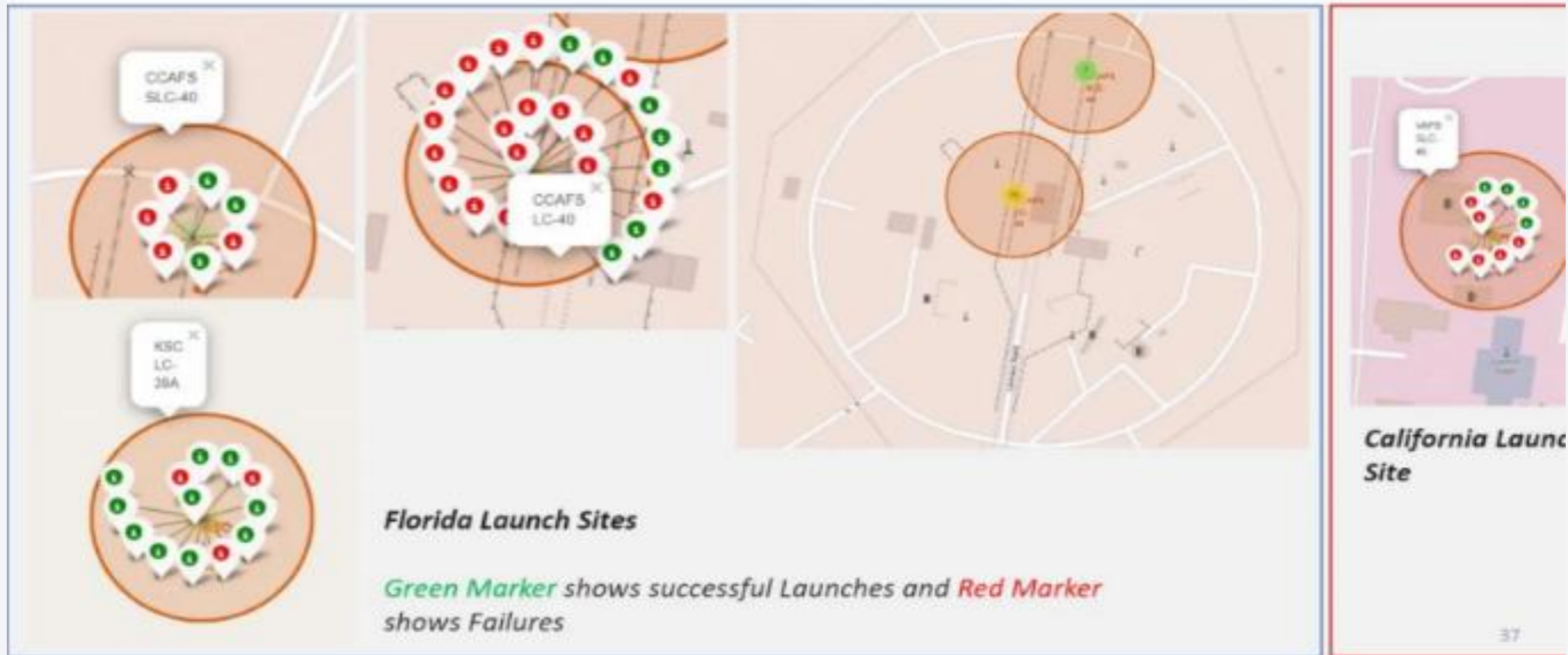
Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

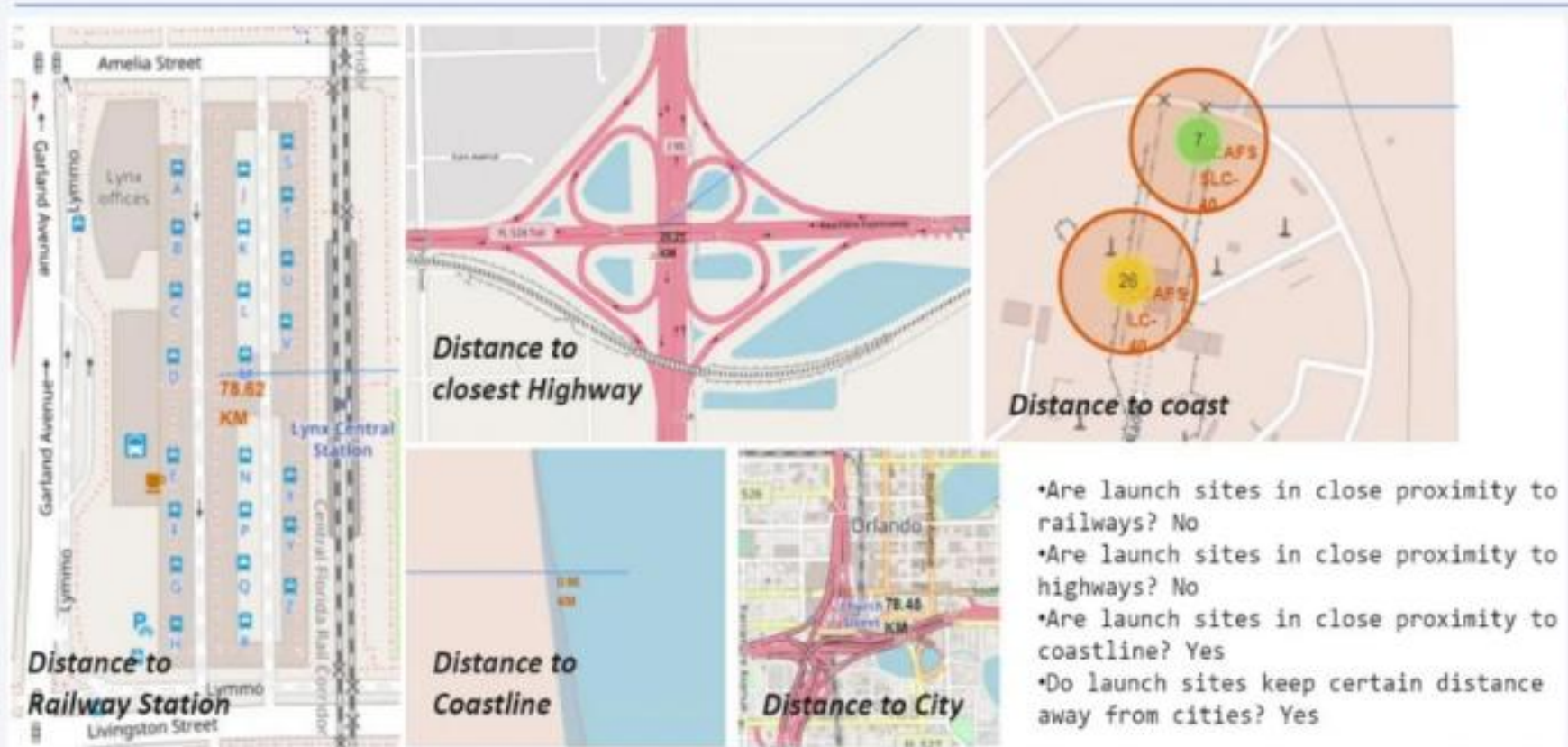
All launch sites global map markers

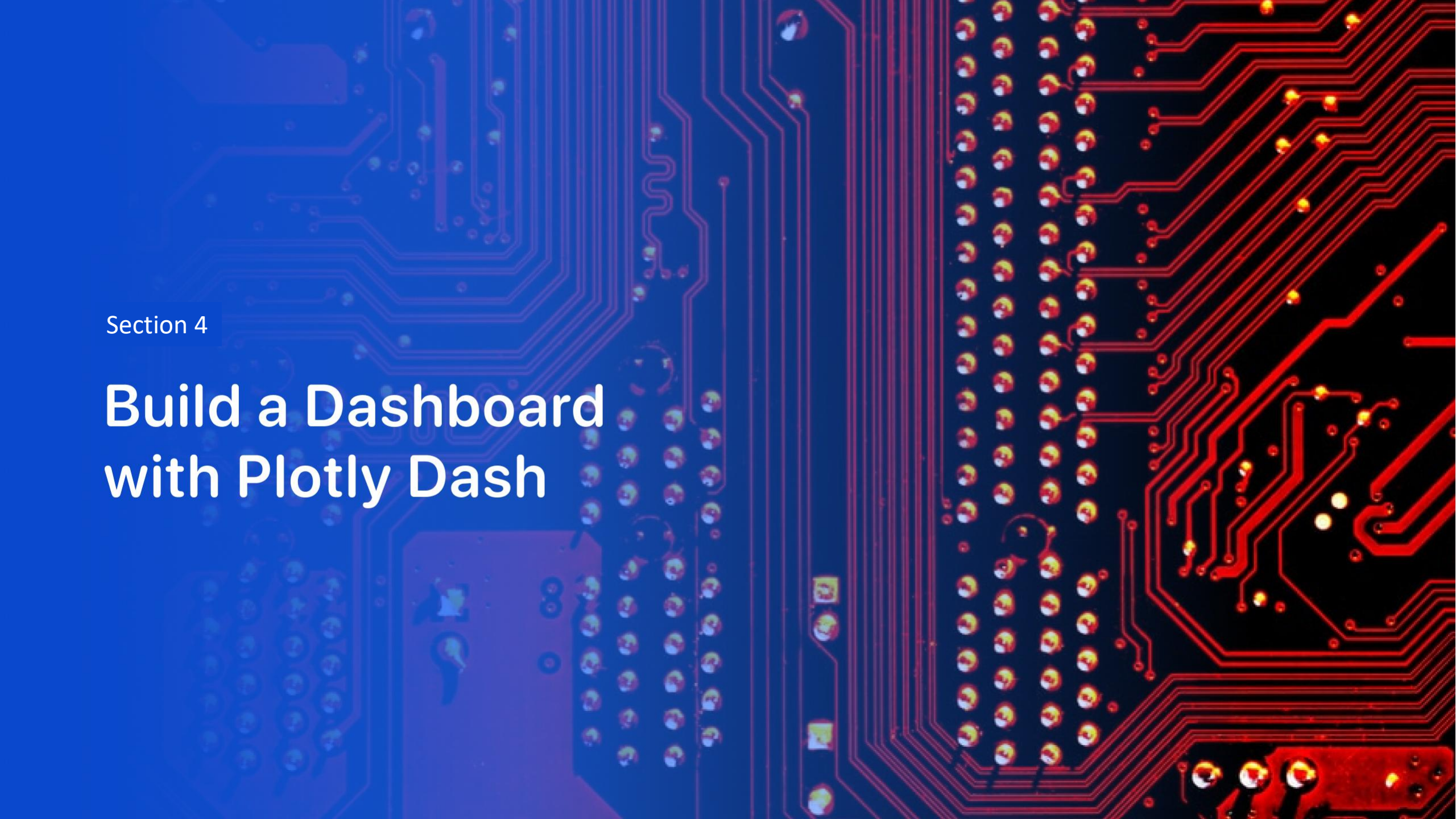


<Folium Map Screenshot 2>



<Folium Map Screenshot 3>





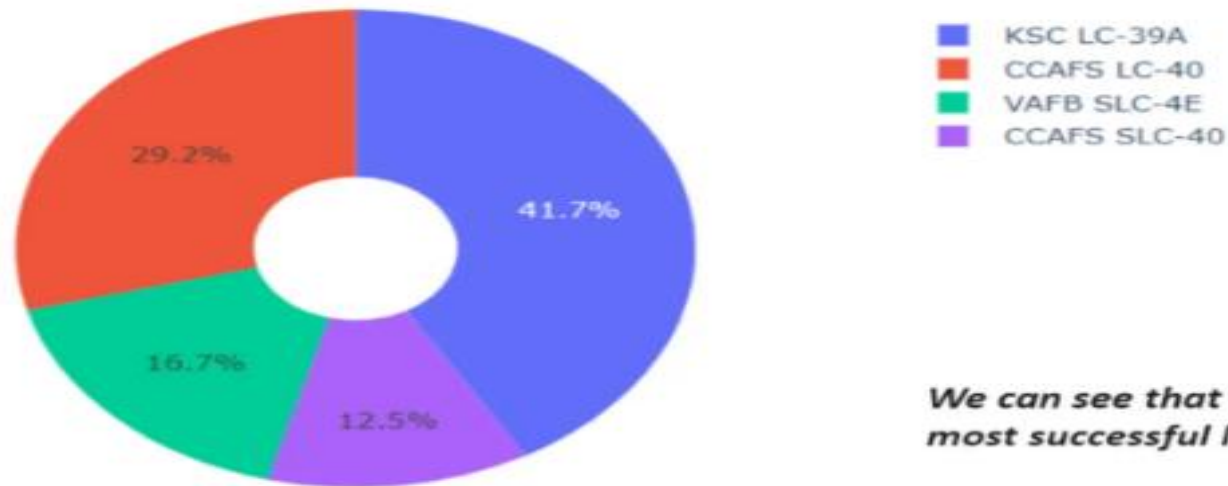
Section 4

Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

Pie chart showing the success percentage achieved by each launch site

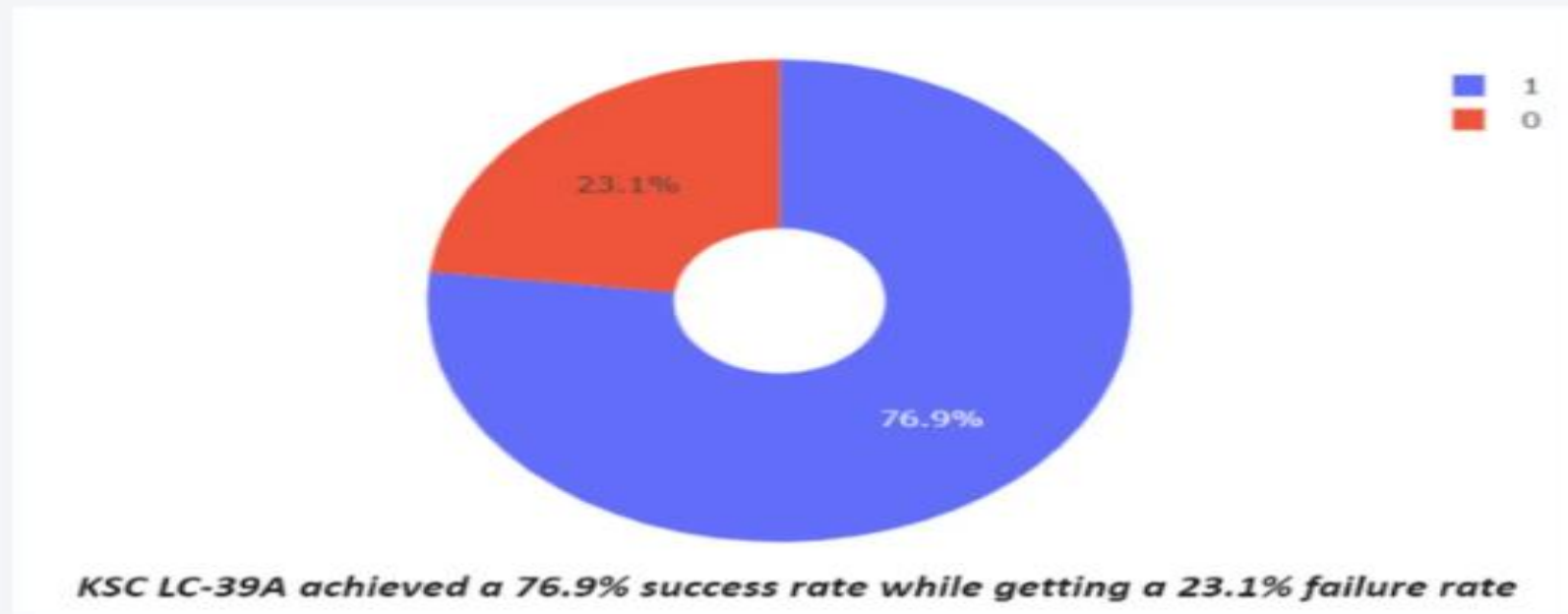
Total Success Launches By all sites



We can see that KSC LC-39A had the most successful launches from all the sites.

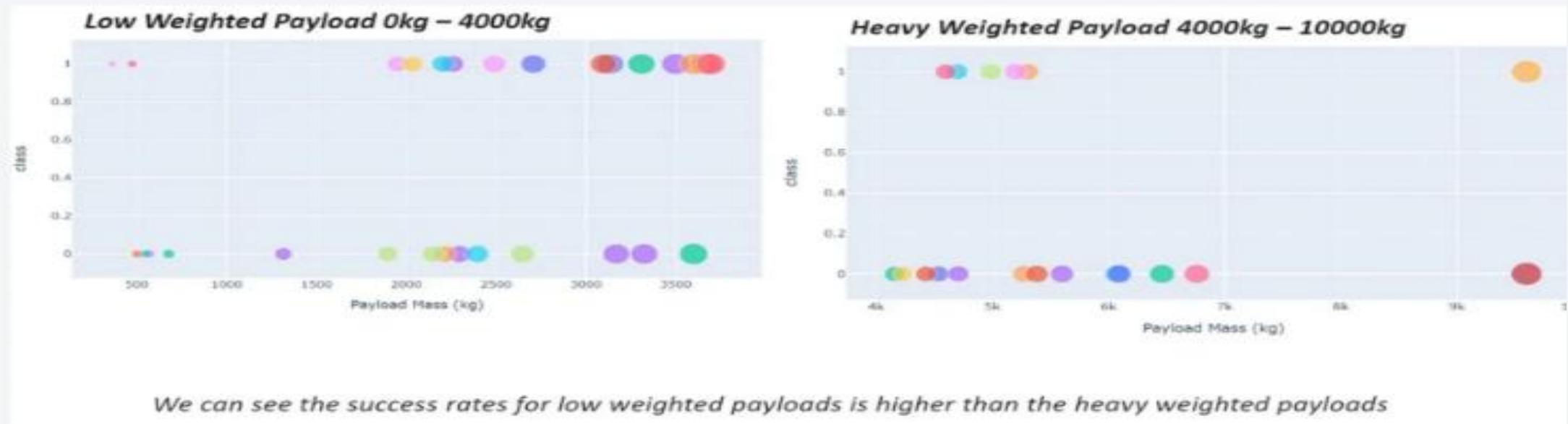
<Dashboard Screenshot 2>

Pie chart showing the Launch site with the highest launch success ratio



<Dashboard Screenshot 3>

Scatter plot of Payload vs Launch Outcome for all sites, with different payload selected in the range slider





Section 5

Predictive Analysis (Classification)

Classification Accuracy

- The decision tree classifier is the model with the highest classification accuracy

```
models = {'KNeighbors': knn_cv.best_score_,
          'DecisionTree': tree_cv.best_score_,
          'LogisticRegression': logreg_cv.best_score_,
          'SupportVector': svm_cv.best_score_}

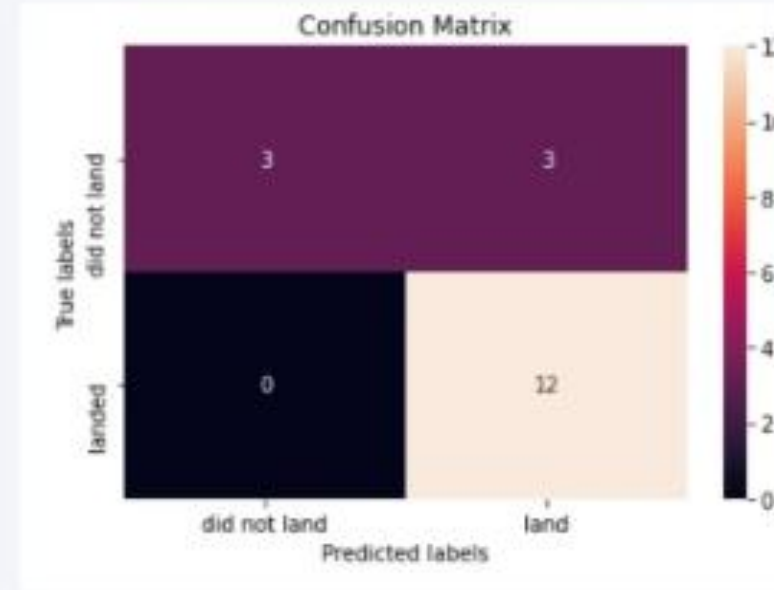
bestalgorithm = max(models, key=models.get)
print('Best model is', bestalgorithm, 'with a score of', models[bestalgorithm])
if bestalgorithm == 'DecisionTree':
    print('Best params is :', tree_cv.best_params_)
if bestalgorithm == 'KNeighbors':
    print('Best params is :', knn_cv.best_params_)
if bestalgorithm == 'LogisticRegression':
    print('Best params is :', logreg_cv.best_params_)
if bestalgorithm == 'SupportVector':
    print('Best params is :', svm_cv.best_params_)
```

Best model is DecisionTree with a score of 0.8732142857142856

Best params is : {'criterion': 'gini', 'max_depth': 6, 'max_features': 'auto', 'min_samples_leaf': 2, 'min_samples_split': 5, 'splitter': 'rand'

Confusion Matrix

- The confusion matrix for the decision tree classifier shows that the classifier can distinguish between the different classes. The major problem is the false positives .i.e., unsuccessful landing marked as successful landing by the classifier.



Conclusions

- SpaceX launch data provides strong predictive insights
- Payload and orbit type significantly affect landing success
- Interactive dashboards improve mission analysis
- Machine learning can effectively predict Falcon 9 landing outcomes
- SVM achieved the best overall classification accuracy

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

